Lipidomic analysis and dengue virus infection

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Summary of main results

Your lipidomic assays went well, all our controls (instrument status, internal standards, blank) were correct.

Context

Dengue fever virus (DENV) is the most widespread arbovirus and a major threat to public health in tropical and subtropical areas of the world. Dengue is arguably the world's most important arboviral disease, with around 390 million infections per year, of which almost 100 million are clinically apparent. While the majority of people infected with DENV recover rapidly, almost 500,000 people a year develop a severe form of the disease, classified as either dengue hemorrhagic fever (DHF) or dengue shock syndrome (DSS). Characterized by increased vascular permeability, hypovolemia and dysregulated blood coagulation, DHF/DSS has a mortality rate of 2.5%1-3. Epidemics of dengue fever (DF) and dengue hemorrhagic fever/dengue shock syndrome (DHF/DSS) have appeared throughout the tropical and subtropical world, with devastating public health consequences for diagnosis and patient care. If the disease is presented early after onset, it is clinically impossible to differentiate between patients infected with DENV, who will experience a mild episode of dengue disease, and those who will progress to potentially fatal DHF/DSS. Moreover, many questions remain unanswered about the molecular mechanisms underlying the infection, and there is currently no specific treatment or vaccine effective against the disease.

Samples

Nature of samples

- Origin(s): Plasma

Received on: 24/09/25

Extracted on: protocol oxylipins and global lipidomic: 28/05/25 and FAME: 30/07/25

Analyzed on: oxylipins: 28/05/25; global lipidomic: 13/06/25 and FAME: 30/07/25

Sample list:

N°ech	Sample list
1	JV-015-RAN-IC-M2
2	BS-007-RAN-01-M3
3	BS-007-RAN-02-M3
4	KT-004-SOC-16-M1
5	BS-135-RAN-IC-M3
6	JV-070-DAV-07-M3
7	BS-129-RAT-10-M5
8	JV-036-SOC-10-M2
9	BS-126-RAT-05-M2
10	JV-017-RAN-10-M3
11	JV-070-DAV-04-M1
12	BS-356-RAN-03-M4
13	JV-090-MIN-IC-M4
17	BS-391-RAN-06-M2
15	JV-061-MIN-03-M5
16	JV-070-MIN-01-M1
17	KT-004-SOC-14-M1
18	JV-089-SOC-04-M1
19	BS-472-RAT-01-M1
20	BS-172-RAT-10-M3

Results

→ Oxylipins results are synthesized in excel file: 250528DMI_Oxylipins_results

Absolute quantification was performed. Values are expressed in pg/µL of plasma.

→ Global results are synthesized in excel file: 250528DMI_Global_Lipidomic_results

Approached quantification was performed. Values are expressed in ng/µL of plasma.

→ FAME results are synthesized in excel file: 250528DMI_AGT_results

Relative quantification was performed. Values are expressed in (metabolite area/ISTD area)/ µL of plasma.

NB: If you use these datas for communication (poster, oral communication, scientific article), please thanks the lipidomic facility core as mentionned:" This work was supported by the MetaboHUB infrastructure funded by the Agence Nationale de la Recherche under the France 2030 program (MetaboHUB ANR-11-INBS-0010 ; MetEx+ ANR-21-ESRE-0035; MetaboHUB (JVCE) ANR-24-INBS-0012).".

Experimental details

Extraction and analysis of global lipidomic

Scope & application

Quantitative analysis of the following classes: Glycerolipids (TG, DG); Cholesteryl ester; Sphingolipids (Ceramids, SM, Mono-hexosylCeramide, Di-hexosylceramide); phospholipids (PC, LPC, PE, LPE, PI, PS, PG); Free fatty acids, and acyl carnitines with internal standard.

Sample preparation

10 µL of plasma were added to a glass tube containing 10 µL of a mixture of internal standards. Lipids were extracted according to a modified Folch extraction (Seyer et Al., Metabolomics, DOI: https://doi.org/10.1007/s11306-016-1023-8, Folch et Al., J. Biol. Chem). A volume sufficient for 100 µL of ultrapure water is added along with 490 µL of a mixture of CHCl3/MeOH (1:1 v/v). The sample was vortexed during 15 minutes. 75 µL of ultra pure H2O is added before vortexing again for one minute and then centrifuging at 2500 rpm for 15 minutes. The upper and the lower phase was then transferred to a glass tube and evaporated under nitrogen and reconstituted in 100µL of a 65:35:5 (v/v/v) MeOH/IPA/H2O mixture.

Analysis

SFC-MS analysis were performed as described (Le Faouder et Al., Metabolites, DOI: 10.3390/metabo11050305). Briefly, lipid mediators were separated on a ACQUITY UPC² Torus diethylamine column (100x3.0 mm, 1.7 µm, Waters) using an Ultra-Performance Convergence Chromatography (UPC²) coupled on-line to an Xevo G2-XS time of flight (Qtof) (Waters, Milford, MA, USA), equipped with an ESI source. The analyses were performed in MS full scan in centroid mode from 50 to 1500 Da, with dynamic range extended (DRE) activated.

Peak integration and quantitative analysis were done using MzMine4.

Experimental details

Extraction and analysis, Total fatty acids

Scope & application

Quantitative analysis of conventional fatty acids: (c10:0, c12:0, c14:0, c15:0, c16:0, c17:0, c18:0, c20:0, c22:0, c23:0, c24:0, c14:1w5, c15:1, c16:1w7, c18:1w9, c18:1w7, c20:1w9, c22:1w9, c24:1w9, c18:2w6, c18:3w6, c18:3w3, c20:2w6, c20:3w3, c20:3w6, c20:4w6, c20:5w3, c22:2w6, c22:6w3, c22:4w6) with internal calibration (internal standard: TG17).

This method allows a relative quantification of the molecules.

Sample preparation:

After global lipidomic analysis, samples were hydrolysed in KOH (0.5 M in methanol, 1mL) at 55°C for 30 min, and transmethylated in boron trifluoride methanol solution 14% (SIGMA, 1 mL) and heptane (1 mL) at 80°C for 1h. After addition of water (1 mL), FAMEs were extracted with heptane (2mL), centrifugate 1min at 2500 rpm and evaporated to dryness and dissolved in ethyl acetate (20 µL).

Total fatty acids' fractions were trans methylated in boron trifluoride methanol solution 14% (SIGMA, 1 mL) and heptane (1 mL) ambient temperature during 5min. After addition of water (1 mL) to the crude, FAMEs were extracted with heptane (2mL), centrifugate 1min at 2500 rpm and evaporated to dryness and dissolved in 15 µL of ethyl acetate.

Analysis:

FAME (1 µL) were analyzed by gas chromatography equipped with a FID (Lillington, J.M., Trafford, D.J., and Makin, H.L. 1981, Clin Chim Acta 111:91-98) on a GC Trace 1600 system using a Famewax RESTEK fused silica capillary columns (30 m x 0.32 mm i.d, 0.25 µm film thickness). Oven temperature was programmed from 100°C to 250°C at a rate of 6°C/min and the carrier gas was hydrogen (1.5mL/min). The injector and the detector were at 220°C and 230°C respectively.

Experimental details

Extraction and analysis of oxylipins

<u>1. Scope & application</u>

Quantitative analysis of the following classes: prostaglandins, thromboxans, leukotriens, lipoxins, resolvins, HODE, HETE, HDoHE, EET, through internal calibration.

<u>Sample preparation:</u>

260 µL of cold methanol and 40 µL of internal standard (Deuterium labeled compounds) were added to 120µL of plasma. After centrifugation at 5000 g for 15 min at 4°C, supernatants were transferred into 2 mL 96-well deep plates and diluted in H2O to 2 mL. Samples were then submitted to solid phase extraction (SPE) using OASIS HLB 96-well plate (30 mg/well, Waters) pretreated with MeOH (1 mL) and equilibrated with 10% MeOH (1 mL). After sample application, extraction plate was washed with 10% MeOH (1 mL). After drying under aspiration, lipids mediators were eluted with 1 mL of MeOH. Prior to LC-MS/MS analysis, samples were evaporated under nitrogen gas and reconstituted in 20 µL on MeOH.

<u>Analysis:</u>

Briefly, lipid mediators were separated on a Kinetex XB-C18 column (2.1 mm, 100 mm, 1.8 µm) (Phenomenex) using Nexera Mikros HPLC system (Shimadzu) coupled to a ESI-triple quadruple 8060 mass spectrometer (Shimadzu). Data were acquired in Dynamic Multiple Reaction Monitoring (DMRM) mode with optimized conditions (ion optics and collision energy). Peak detection, integration and quantitative analysis were done using LabSolution analysis software (Shimadzu) based on calibration lines built with commercially available oxylipins standards (Cayman Chemicals).

METABOHUB, the National infrastructure in metabolomics & fluxomics

MetaboHUB is the French Infrastructure in Metabolomics & Fluxomics created in 2013 in the framework of the Program “Investissements d’Avenir” (Investment for the Future) launched by the French Ministry of Research, Higher Education and Innovation (MESRI) and the National Agency for Research (ANR).
MetaboHUB addresses key critical priorities in metabolomics, fluxomics and bioinformatics to (i) provide high throughput quantitative analytical technologies for biochemical phenotyping of large variety of biological samples, (ii) identify metabolites and annotate metabolomes, (iii) develop broadband flux measurements, (iv) provide access to high value-added services to national scientific community and industry players, (v) attract and train a new generation of scientists and users.